SENIOR HIGH SCHOOL • COMPUTER PROGRAMMING (.NET TECHNOLOGY)

The CSS Box Model

Lesson 5

Grade 11 | Quarter 2 | CPNT 2.3

LESSON

5

The CSS Box Model

Structuring layouts with content, padding, border, and margin

CPNT 2.3

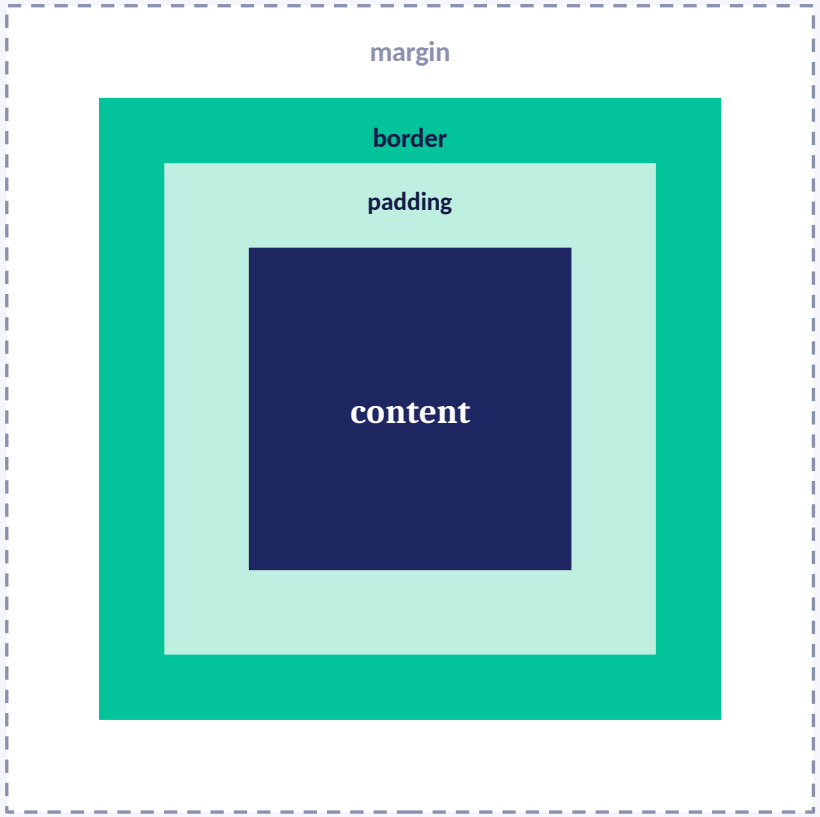
What You Will Be Able To Do

Unpacked Performance Standards

1. Explain the components of the CSS box model.
2. Apply margin, border, and padding properties in webpage layouts.
3. Design organized webpage layouts using the box model.

The Box Model: Four Layers

Every HTML element is a rectangular box made of four layers, nested from the inside out.



Content

The actual text, image, or element itself. Its size is set by width and height.

Padding

Clear space between the content and the border — like cushioning inside a package.

Border

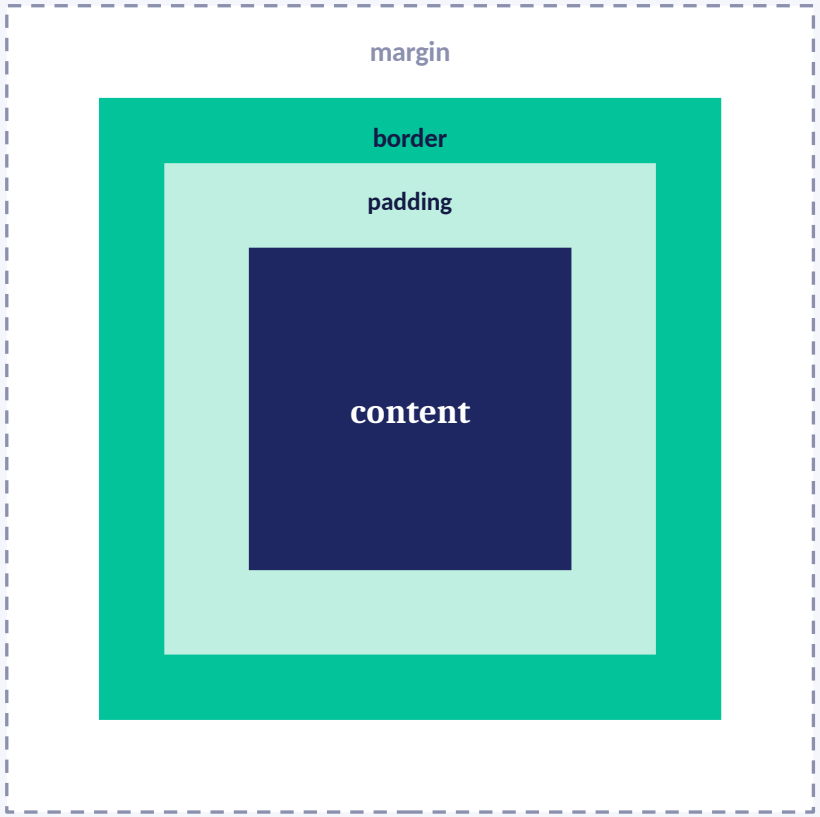
A visible (or invisible) line that wraps around the padding and content.

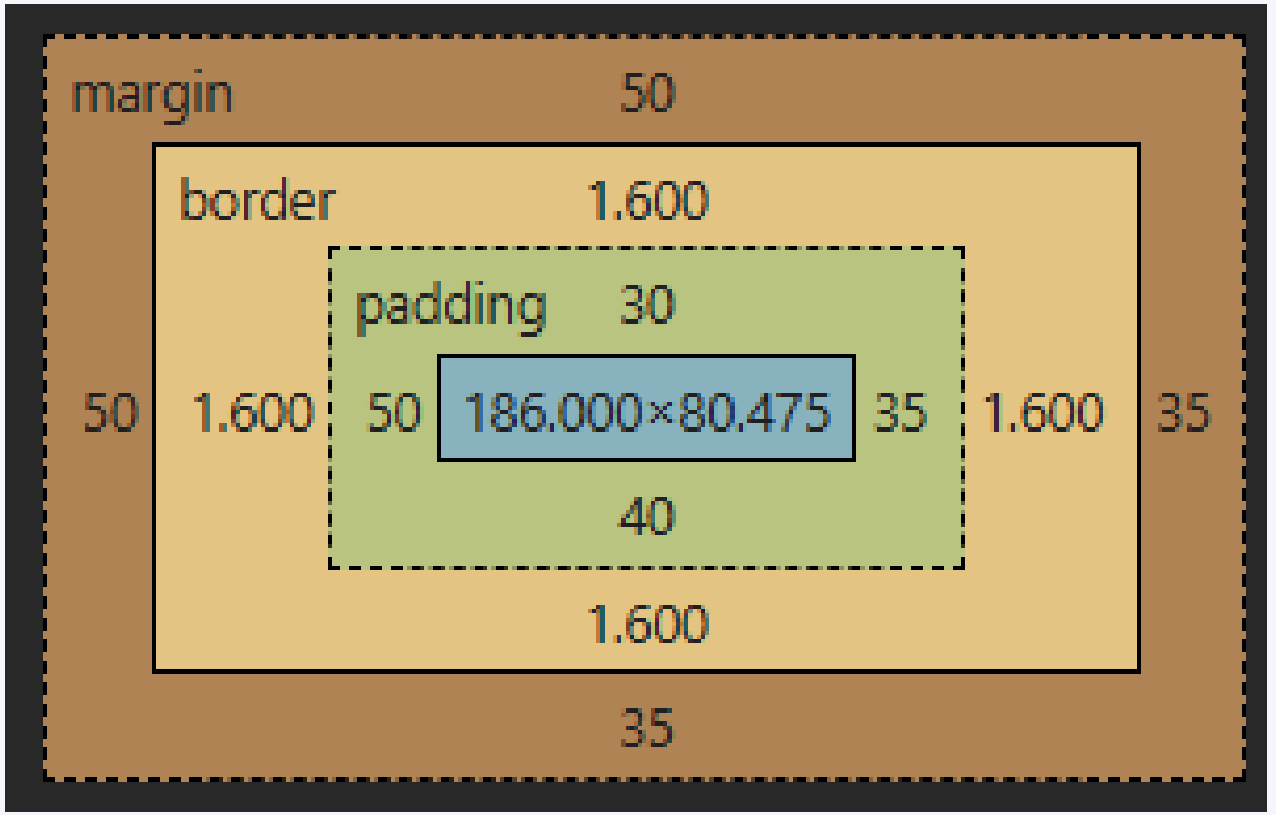
Margin

Space outside the border that pushes neighboring elements away.

The Box Model: Four Layers

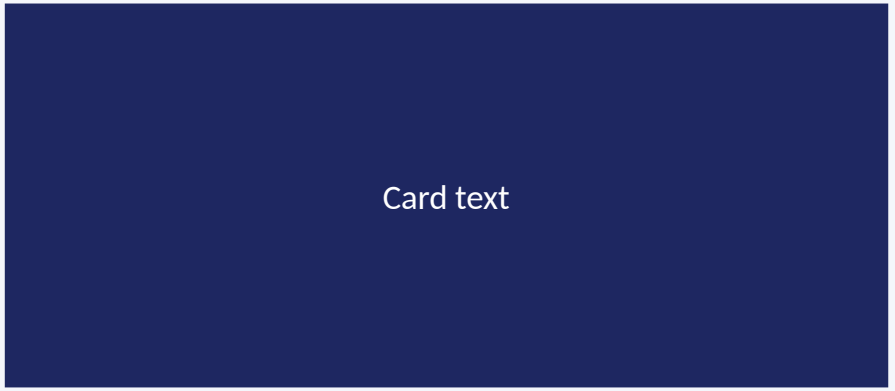
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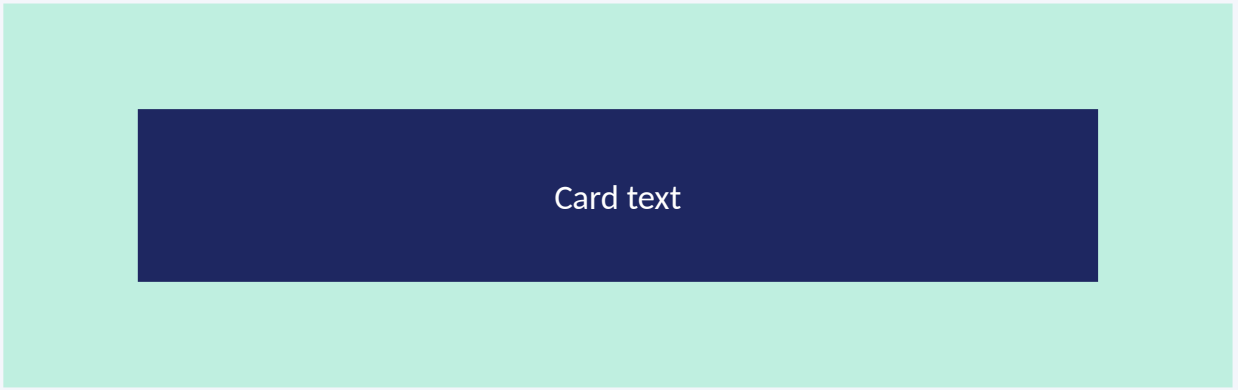


Example 1: Padding — Space Inside the Box

Padding pushes content away from its own border, without affecting neighboring elements.



padding: 0



padding: 24px

```
.card {
  padding: 24px;
}
```

Onsite: teacher toggles padding live in dev tools. Online: students compare both cards in a browser.

Example 2: Border — Framing the Box

A border wraps around the padding and content, giving the element a visible edge.



`border: none` `border: 4px solid mint`

```
.card {
  border: 4px solid #02C39A;
}
```

Onsite: teacher live-codes different border-styles. Online: students try solid, dashed, and dotted borders.

Example 3: Margin — Space Outside the Box

Margin pushes an element away from its neighbors, creating breathing room between elements.



margin: 0



margin: 24px

```
.card {
  margin: 0 24px;
}
```

Onsite: teacher demonstrates spacing between multiple cards. Online: students adjust margin values live.

Example 4: box-sizing — Content-box vs. Border-box

Same declared width, padding, and border — but box-sizing changes the final rendered size.

content-box (default)



Rendered total: 250px wide

border-box



Rendered total: 200px wide

```
.content-box { box-sizing: content-box; }  /* padding + border ADD to width */
.border-box { box-sizing: border-box; }  /* padding + border INCLUDED in width */
```

Onsite: teacher resizes both boxes live in dev tools. Online: students inspect computed widths in the browser.

LESSON 5 • ASSESSMENT

Quick Check: The Box Model

Formative assessment — short quiz (identification / short answer), answered individually before the layout activity.

1. List the four layers of the CSS box model, from innermost to outermost.

2. Which property adds space INSIDE an element, between its content and border?

3. Which property adds space OUTSIDE an element, separating it from its neighbors?

4. What does box-sizing: border-box change about how width is calculated?

5. A box has width: 200px, padding: 10px, and border: 2px, using content-box. What is its total rendered width?

LESSON 5 • ACTIVITY

Hands-On: Design a Card Layout

Written Work: Box Model Layout Exercise

Apply padding, border, and margin to organize a set of content cards.

1. Open the provided webpage with a set of plain, unstyled content cards.
2. Add padding inside each card so the text doesn't touch the edges.
3. Add a border around each card to visually separate it from the background.
4. Add margin between cards so they don't touch each other, then preview the layout.

NEXT UP

Lesson 6: CSS List and Links

With layouts under control, we'll style lists and hyperlinks in Quarter 2's CPNT 2.4 — which also kicks off Performance Task 1.

Thank you!